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Engineering, Built Environment and IT
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Tutorial 7, Backprop, Hopfield, ID3 and K-Means COS 314

Instructions

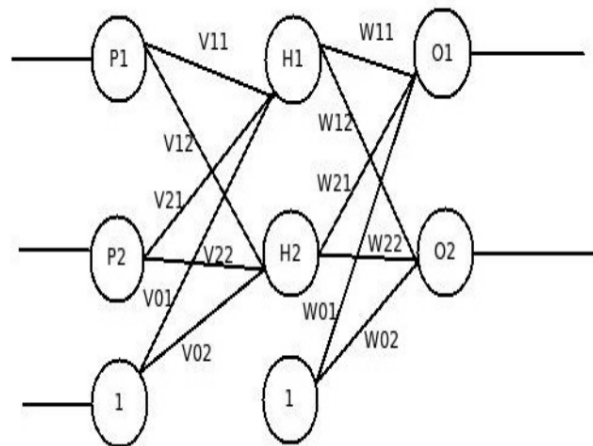


Figure 1: Network

- $V_{01} = 0.26, V_{02} = -0.15, V_{11} = 0.3, V_{12} = 0.5, V_{21} = -0.25, V_{22} = 0.5, W_{11} = 0.15, W_{12} = 0.35, W_{21} = -0.05, W_{22} = 0.55, W_{01} = 0.12, W_{02} = 0.13, t_1 = 0.2, t_2 = 0.8, P_1 = 0.68, P_2 = 0.35$ answer the questions that follow.
 - Evaluate the values of H_1, H_2, O_1 and O_2 in the forward pass for the following inputs $P_1 = 0.68$ and $P_2 = 0.35$.
 $H_1 = \mathbf{0.593}, H_2 = \mathbf{0.589}, O_1 = \mathbf{0.545}$ and $O_2 = \mathbf{0.66}$
 - Calculate δ_1 and δ_2 for the inputs in (a)
 $\delta_1 = \mathbf{-0.09}$ and $\delta_2 = \mathbf{0.03}$
 - Update weight W_{11} and V_{11}
 $\mathbf{W_{11} = 0.1}$ and $\mathbf{V_{11} = 0.3}$
- Given that a Hopfield neural network is used to store the following pattern $[1 \ 0 \ 1 \ 0]$, illustrate how this network would be able to perform autoassociation given the following input $[1 \ 1 \ 1 \ 0]$.
Use the same format as in the notes (and slides) to work out the steps of autocorrection.
- Given the following dataset, which relates to the classification of whether an individual qualifies for contact lenses or not. Use the Information Gain Decision Tree induction method to determine which parameter will be the root node of the decision tree.
Age = 1 – young, 2 – middle, 3 – elderly, **Vision** = 1 – short, 2 – long, **Astigmatic** = 1 – no, 2 – yes, **Tears** = 1 – reduced, 2 – normal, **Class (contact lenses)** = 1 – hard, 2 – soft, 3 – none

Age	Vision	Astig	Tears	Class
1	1	1	1	3
1	1	1	2	2
1	1	2	1	3
1	1	2	2	1
1	2	1	1	3
1	2	1	2	2
1	2	2	1	3
1	2	2	2	1
2	1	1	1	3
2	1	1	2	2
2	1	2	1	3
2	2	2	2	3
3	1	1	1	3
3	1	2	1	3
3	2	1	1	3
3	2	1	2	2
3	2	2	2	3
3	1	2	2	1

Table 1: Contact Lenses Data

Table 2: Information Gain Comparison for Root Node Selection

Parameter	Information Gain
Tears	0.5822
Astigmatic	0.2125
Age	0.0833
Vision	0.0392

4. Given the Table 3 measurements of 16 objects. Using the following values as the initial centroids: C1 - (3.8,9.9), C2-(7.8,12.2) and C3 (6.2, 18.5). Perform three iterations of the K-means algorithm and show the centroid updates at each iteration.

Table 3: Object Metrics Dataset

x	6.8	0.8	1.2	2.8	3.8	4.4	4.8	6.0	6.2	7.6	7.8	6.6	8.2	8.4	9.0	9.6
y	12.6	9.8	11.6	9.6	9.9	6.5	1.1	19.9	18.5	17.4	12.2	7.7	4.5	6.9	3.4	11.1

Table 4: K-Means Centroid Updates Over Three Iterations

Centroid	Initial State	End of Iteration 1	End of Iteration 2	End of Iteration 3
C1	(3.8, 9.9)	(2.97, 8.08)	(2.97, 8.08)	(2.97, 8.08)
C2	(7.8, 12.2)	(8.06, 8.34)	(8.06, 8.34)	(8.06, 8.34)
C3	(6.2, 18.5)	(6.60, 18.60)	(6.60, 18.60)	(6.60, 18.60)